

Chapter 4. Secure Channel

4.1. General Description

The Secure Channel and Message Layer provides a consistent networking service substrate to allow Nodes to communicate securely with one another.

During commissioning and unicast communication, a [discovery](#) mechanism is provided to determine peer IPv6 addresses and operational parameters. [Secure session establishment](#) mechanisms are provided using either [certificates \(CASE\)](#) or [shared passcodes \(PASE\)](#).

4.1.1. Messages

Communication is performed using messages. Messages are either secured or unsecured.

Each message has a [Session Type](#) and [Session ID](#) in order to identify whether it is secured and how it is to be decrypted and authenticated if it is. Each message has a [Message Counter](#) field in order to uniquely identify the message for the purposes of security and duplicate detection.

Operational communication is defined as traffic that uses the secured Matter message format between commissioned nodes over an IP transport. All operational communication has [message security](#) enabled. Operational communication between Nodes can be either unicast or multicast.

Unsecured communication is strictly limited to:

- [Discovery](#), which does not use the Matter message format.
- [User Directed Commissioning \(UDC\)](#), which uses unsecured messages to initiate the commissioning process.
- [Session establishment](#), which uses unsecured messages to establish a [CASE](#) or [PASE](#) session.

4.1.1.1. Message Types

Messages are defined as either a *control message* or *data message*. Most messages are data messages. Control messages are reserved for internal protocols such as [MCSP](#) to initialize security. Both message types are identical in format, but use separate message counter domains so they can operate securely over the same security key.

4.1.1.2. Message Transports

Messages are of [finite size](#) and are sent as individual packets over the supported transports:

- UDP transports each message as a separate datagram. Messages support a basic reliability protocol called [MRP](#) for use when the underlying transport (in this case UDP) doesn't provide such features.
- TCP transports each message with a length prepended, performing segmentation and reassembly as needed.
- [BTP](#) transports each message over BLE as a separate SDU, performing segmentation and